

Alessio Salvatore

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Education

ETH Zürich, Zürich, Switzerland Sep 2025–Present
M.Sc. in Robotics

Coursework: Probabilistic Artificial Intelligence, Advanced Machine Learning, Image Analysis and Computer Vision, Planning and Decision Making for Autonomous Robots, 3D Vision, Perception and Learning for Robotics, Deep Generative Models, Robot Learning

Politecnico di Milano, Milan, Italy Sep 2022–Jul 2025
B.Sc. in Automation Engineering

Final Grade: 110/110 *cum laude* (equivalent GPA $\approx$ 4.0/4.0; 29.57/30).

Illinois Institute of Technology, Chicago, IL, USA Aug–Dec 2024
Exchange — Global E3 Program

Projects

Research Projects

Knowledge Distillation for Aerial Robots Control  Video  Report Feb–Jun 2026
Autonomous Systems Lab – Roland Siegwart Lab

- Multi-teacher distillation for the OMAV, an overactuated tilt-rotor MAV: each of 30,000 randomized simulated vehicles is flown by its own parameter-matched model-based expert, and the full family of experts is compressed into a single LSTM policy – trained closed-loop by backpropagating through the differentiable vehicle dynamics (BPTT, $\sim$ 42 h of simulated flight)
- Outperforms the nominal model-based controller in velocity tracking (2,000 paired rollouts) and nearly halves its divergence rate under $\pm 30\%$ model mismatch, approaching the parameter-matched teachers; deployed zero-shot on the real OMAV

Goal-Driven Adaptive-Resolution 3D Mapping with RL  Report Feb–Jun 2026
Computer Vision and Geometry Group – Marc Pollefeys Lab

- Dense 3D maps waste memory on detail the task never uses – trained a PPO agent that decides, region by region and online during navigation, where the map needs fine voxels and where coarse ones suffice, under a hard memory budget
- On unseen scenes the learned map keeps 79% of point-goal navigation success with 87% less memory than a uniformly fine map (20-episode validation); built as a sparse hierarchical multi-resolution rewrite of PEANUT's voxel map (Habitat, HM3D)

Selected Projects

Autonomous Exploration on a Unitree A2 Quadruped  Repo 06/2026
ETH Robotics Summer School (ETHZ-RobotX)

- Our quadruped autonomously searched three real field sites (Hesai LiDAR; TARE/FAR planners, DLIO LiDAR-inertial odometry) and reported the objects it found
- Built the mission's scored output: the artifact-mapping node fusing YOLO detections with TF2 transforms into a clean, deduplicated map of object positions

Vision-Language-Action Policies on a Real SO-101 Arm  Repo 05/2026
Project of the Class Robot Learning

- Fine-tuned SmoVLA policies to follow written commands – “place the can on the person left of Obama” – across three pick-and-place tasks, generalizing to celebrity faces never seen in training; on-robot inference at 10 FPS
- Owned the data side: LeRobot pipeline with prompt relabeling, brightness and face-swap augmentation, DAgger-style corrective demonstrations, and automated H100 cloud training with checkpoint streaming to the Hugging Face Hub

Person Following on the Agibot X2 Humanoid  Repo 05/2026
RoboHack Hackathon

- In one hackathon weekend, made the humanoid find a person and follow them at a fixed distance with everything running onboard: YOLOv8 on the head stereo cameras with disparity-based depth, Ruckig head tracking, and a state-machine locomotion supervisor

Channel Position Estimation from Pipe-Inspection Video Repo

05/2026

Physical AI Hackathon

- Recovered where a sewer-inspection probe is along the pipe at every frame – plus when it turned around – from its video alone: no encoder, no learning, just sparse optical flow with outlier filtering, scene gating, and scale-free normalization; ~1000 frames in under 8 s on CPU

DINOv3 for Mitral Valve Segmentation Repo

12/2025

- Clinicians annotate only a few frames per medical video – propagated those sparse labels through whole videos via DINO-based feature propagation to generate a full pseudo-labeled training set, then trained a U-Net segmentation model on it

Multi-Agent Multi-Goal Collection Planner

12/2025

Project of the Class Planning and Decision Making for Autonomous Robots

- A fleet of robots divides up collection goals among themselves (Large Neighborhood Search), plans paths in real time, and avoids each other with no central coordinator – each robot seeing only part of the map

Autonomous Satellite Docking Planner

11/2025 – 12/2025

Project of the Class Planning and Decision Making for Autonomous Robots

- Planned collision-free docking trajectories for a satellite maneuvering through orbital clutter, solving the non-convex guidance problem as a sequence of convex ones (Sequential Convexification)

Work Experience

JOINT Lab — Italian Institute of Technology, Bergamo

Feb–Jun 2025

Research Intern — Industrial Robotics  Repo

- Took a linear actuator from CAD design (commercial off-the-shelf components) to real-time control on Arduino, with a MATLAB simulation validating the system dynamics

ETH Robotics Club

Nov 2025–Present

Partnerships Team Lead

- Secured industry sponsorships for RoboHack 2026, the club's flagship robotics hackathon

Laelaps AI

Jun 2026–Present

Robotics Engineer

- Developing a multi-agent reinforcement learning (MARL) framework for efficient, scalable patrolling of large areas with teams of robots